prone to develop T2D later in life.

The medical practitioners and healthcare researchers have taken various steps to tackle the problem of diabetes, such as the development of a mathematical model of glucose-insulin dynamics, automatic infusion pumps, sensors to monitor glucose level, and a control system for automatic infusion of insulin.

By using new technologies in glucose monitoring and insulin infusion rate, it is possible to sense the glucose level by using continuous measurements of glucose monitoring. The industrial and scientific community has been increasing its efforts to develop and replicate the technological phenomenon of the pancreas through artificial pancreas (AP) to control insulin delivery.

What is artificial pancreas?

Artificial pancreas is a man-made scientific technology developed in order to match the working of pancreas. It is designed to change glucose levels in the bloodstream in a similar way as the human pancreas does throughout the day and overnight.

Maintaining a balanced glucose level is important for the proper functioning of brain, kidney, and liver. Therefore it is important for T1D patients to maintain these levels when the body cannot produce insulin.

An AP system consists of three devices—an insulin delivery pump, a continuous glucose monitoring system (CGM), and a computer-controlled algorithm in order to allow real-time communication between two devices. The AP system is sometimes referred to as a closed-loop device as the patient is not in the decision-making loop, or an automatic system for glucose control.

After monitoring the blood glucose levels, the AP system manipulates insulin infusion pump rates by a closed-loop controller that receives information from the sensor in order to reduce the incidence of low blood glucose due to the over administration of insulin (hypoglycemia) and high blood sugar due to failure of enough administration of insulin or high-intensity exercise (hyperglycemia). Hypoglycemia is a short-term risk that results in drowsiness, shakiness, and even loss of consciousness. Hyperglycemia is a long-term risk that results in blindness (diabetic retinopathy), numbness (diabetic neuropathy), and kidney failure (diabetic nephropathy).

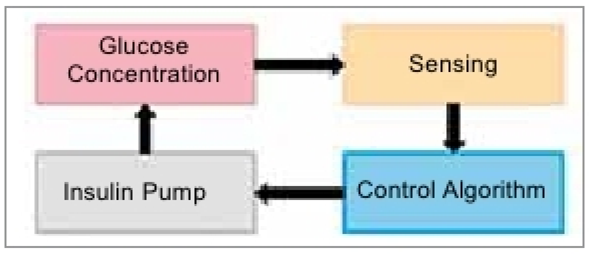

Fig. 2: Closed-loop process of artificial pancreas

Researchers are currently working on three main types of artificial pancreas:

1. Closed-loop artificial pancreas. Closed-loop artificial pancreas is also called a closed-loop insulin system in which an insulin pump communicates wirelessly with a CGM inserted under the skin. The CGM measures blood sugar concentrations in patient cells and sends the result to a small computer where the control algorithm analyses the result and calculates the correct insulin dosage.

2. Bionic pancreas. Bionic pancreas is being developed by Dr Edward Damiano's Beta Bionics firm. It consists of two pumps that deliver insulin and glucagon, respectively, and automatically controls blood glucose levels. The pump is wirelessly connected to the iPhone that enables real-time communication between devices and calculates the required insulin or glucagon doses.

3. Implanted AP. Developed by researchers of De Montfort University, the implanted pancreas contains a gel that acts according to the changes in glucose level. The gel administers a higher dose of insulin if the concentration of glucose increases and decreases the amount of insulin during low glucose concentration. It can be refilled with insulin consistently.

According to researchers, the closed-loop system of blood glucose regulation reduces the risk of hypoglycemia in adults.

How artificial pancreas works?

The complete AP model is shown in Fig. 1, with its major operation described in Fig. 2. The AP comprises following units:

CGM. A CGM takes ongoing blood glucose readings through a little sensor inserted into the skin and maintains a stable flow of information about diabetic patient glucose levels in the bloodstream. A sensor that is fitted under the patient's skin (subcutaneously) continuously monitors the concentration of glucose in the blood around cells. A small transmitter sends data to the receiver. CGM provides a continuous display of estimates of both blood glucose levels and direction and rate of change of these estimates. To get the correct predictions of blood glucose from a CGM, the diabetic infected patient needs to calibrate the CGM periodically using measurements of glucose from a blood glucose device.

Control algorithm. A computer model or controlled algorithm embedded in an external processor, also called the controller, performs a series of mathematical calculations after receiving information from a CGM. The controller manipulates the insulin infusion rate based on these calculations.

Insulin pump. It injects the correct dosage of insulin to the fatty tissue below the skin according to the instructions sent by the control algorithm. As a result, insulin moves throughout the bloodstream, thus lowering blood glucose levels.

Patients. Patients are a significant part of the AP system. The amount of glucose in the bloodstream frequently changes as it gets affected by food taken by the patient, intensity of physical activity, and other substances.

Current international trends

Artificial pancreas device based on technology from the University of Virginia, Center for Diabetes Technology has been accepted by the US FDA (Food and Drug Administration). This device, also called Control-IQ, automatically injects the required dosage of insulin to patients with T1D. It is manufactured by Tandem Diabetes Care that monitors glucose levels with Dexcom G6 CGM. People using this system are free from checking blood glucose levels frequently in a day by fingerstick and also from the delivery of insulin by injections multiples time a day.

The Omnipod Horizon System announced the successful trials of their AP system that uses a tubeless patch pump instead of a regular tubed one. It is the most recent Dexcom CGM technology, an improved version of insu-